

Social Network Analysis of the MUSAE GitHub Developer Collaboration Network

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Abstract

This study presents a comprehensive, multi-layered social network analysis of the MUSAE GitHub developer collaboration network, extending the macro-level small-world findings of Zhang et al. (2025) to the micro-level of 37,700 individual developers and 289,003 mutual follow relationships. At the structural level, we validate the small-world phenomenon at the developer scale, yielding a sigma value of $\sigma = 935.55$, exceeding Zhang et al.'s economy-level index (4.62–5.84) by two orders of magnitude, driven by hyper-connected Web developer hubs bridging segregated technical communities. Community detection via the Louvain algorithm reveals 31 homophilic communities ($Q = 0.4534$), with ML and Web developers self-organizing into structurally distinct clusters. Link prediction experiments, conducted with strict data leakage prevention on a held-out 10% edge test set, demonstrate that the Resource Allocation index achieves the highest predictive accuracy ($AUC = 0.8474$), with Web-to-Web collaboration being substantially more predictable than ML-to-ML ($AUC = 0.860$ vs. 0.764), independently corroborating the homophily finding. Information diffusion simulations using Independent Cascade and Linear Threshold models reveal that viral saturation is achievable from a single seed node (44.3% network reach), while consensus-driven adoption scales progressively with seed set size (15.2% to 68.7%). To provide temporal validation, we construct an independent timestamped interaction graph from GH Archive event data (June 1–7, 2019), enabling temporal link prediction with a time-based train/test split, daily network evolution tracking, and burst activity detection. The best-performing temporal link prediction AUC is **0.9904**, compared to **0.8474**, providing cross-dataset validation of the structural findings. Together, these analyses provide the first developer-level temporal and structural characterization of the GitHub follow network, with practical implications for open-source tool dissemination and community-targeted adoption strategies.

Keywords: Social Network Analysis, GitHub, Small-World, Community Detection, Link Prediction, Information Diffusion, PageRank, Independent Cascade, Linear Threshold

1. Introduction

The structure of open-source collaboration has significant implications for how knowledge, tools, and best practices spread across the software engineering ecosystem. Recent work by Zhang et al. (2025) validated the small-world phenomenon within GitHub at an economy and country-aggregate level, demonstrating that international collaborations form tightly knit clusters. However, their study aggregated data geographically and did not perform developer-level structural analysis, link prediction, or simulate information diffusion.

This study bridges that gap by analyzing the fine-grained developer follow-network. We aim to answer three specific research questions:

RQ1: What is the community structure of the developer network, and does it align with developer specializations (ML vs. Web)?

RQ2: Who are the most influential developers in the network according to centrality and PageRank metrics, and which domain do they represent?

RQ3: Can structural similarity metrics effectively predict future or hidden collaborations without data leakage?

To address the static snapshot limitation inherent in the MUSAE dataset, this study further constructs an independent temporal interaction graph from GH Archive event data covering the MUSAE collection period (June 2019), enabling a fourth research question: **RQ4:** How does the GitHub developer interaction network evolve over time, and does temporal link prediction validate the structural findings from the static MUSAE analysis?

2. Related Work

The foundational motivation for this study stems from Zhang et al. (2025), who investigated the macro-level small-world characteristics of global open-source software collaboration. At the micro-level, the MUSAE dataset utilized in our research was originally introduced by Rozemberczki et al. (2021) to evaluate node embedding and graph neural network techniques on multi-attributed networks.

Beyond structural descriptions, predicting future interactions is a core task in network science. Liben-Nowell and Kleinberg (2007) pioneered the use of topological similarity metrics (such as Adamic-Adar and Jaccard coefficients) for link prediction in co-authorship networks, which closely mirror open-source developer networks. Furthermore, the mechanics of viral information spread have been extensively modeled by Kempe, Kleinberg, and Tardos (2003) using the Independent Cascade (IC) and Linear Threshold (LT) models, providing the theoretical framework for our influence diffusion simulations. The use of timestamped event logs for temporal network analysis has a strong precedent in software engineering research. Clauset et al. demonstrated that temporal network snapshots reveal structural dynamics invisible in static aggregations, a finding directly motivating our GH Archive extension. GH Archive, which records the public GitHub event timeline since 2011, has been used to study temporal collaboration patterns at scale, providing the empirical foundation for our temporal analysis component.

3. Dataset

This project utilizes the MUSAE GitHub Social Network dataset [2]. Nodes represent GitHub developers who have starred at least 10 repositories, and undirected edges represent mutual follower relationships. Each node is annotated with a binary target label indicating the developer's primary domain: 0 for Web developers and 1 for Machine Learning (ML) developers. The exact structural properties of the network are summarized in Table 1.

Table 1: Network Statistics

Statistic	Value	Scope
Number of nodes	37,700	Full graph
Number of edges	289,003	Full graph
Graph density	0.000407	Full graph
Avg clustering coefficient	0.1675	Full graph
Number of connected components	1	Full graph
Largest component size (LCC)	37,700	LCC
Approx. diameter (LCC)	11.0	LCC (approximation)
Avg path length	4.8384	LCC sample (n=5000, seed=42)

Statistic	Value	Scope
Degree assortativity	-0.0752	Full graph
Small-world sigma (σ)	935.555	LCC vs ER random

4. Methodology

4.1 Part A: Network Statistics & Small-World Validation

We instantiated the network as an undirected graph using NetworkX [3]. To validate the small-world phenomenon, we computed the sigma coefficient (σ) by comparing the empirical network's average clustering and path length against an equivalent Erdos-Renyi (ER) random graph of identical size and density. Due to computational constraints on calculating exact shortest paths for 37,700 nodes, the average path length was approximated using a robust 5,000-node random sample (seed=42). A value of $\sigma > 1$ confirms the small-world hypothesis.

4.2 Part B: Centrality & Influence

We computed four standard centrality measures: Degree, Closeness, Eigenvector, and Betweenness centrality. Betweenness centrality requires all-pairs shortest paths, which is computationally prohibitive on large graphs; thus, it was approximated using $k=500$ pivot nodes (seed=42). Standard PageRank and Personalized PageRank (PPR) vectors were also calculated, with PPR seeded specifically on ML nodes and Web nodes separately to identify domain-specific influence cores.

4.3 Part C: Community Detection

The Louvain heuristic method was employed to extract the community structure by maximizing the modularity score Q . This greedy algorithm is well-suited to large sparse networks and produces hierarchically nested communities. We subsequently mapped the binary developer type labels back to the identified communities to analyze homophily — the tendency of similar nodes to cluster together.

4.4 Part D: Link Prediction & Information Diffusion

We evaluated four topological similarity metrics for link prediction: Common Neighbors, Jaccard Coefficient, Adamic-Adar Index, and Resource Allocation Index. To rigorously prevent data leakage, 10% of true edges were masked out into a test set alongside an equal number of true non-edges sampled from non-edges of the training graph only. All four similarity scores were computed strictly on the 90% training graph G_{train} , never on the full graph G_{lcc} . Performance was evaluated using AUC-ROC, Average Precision, and Precision@ k ($k = 10, 50, 100$).

For information diffusion (Part D2), we simulated the Independent Cascade (IC) and Linear Threshold (LT) models [5] over the entire network. Seed sets of varying sizes ($k \in \{1, 5, 10, 25, 50\}$) were selected using five different centrality-based strategies: PageRank, PPR-ML, PPR-Web, Degree, and Betweenness. The IC model caps propagation at 20 rounds to reflect realistic diffusion horizons; the LT model runs until convergence with thresholds drawn from Uniform(0,1) per simulation.

4.5 Part E: Temporal Analysis

To complement the static MUSAE dataset, an independent temporal interaction graph was constructed using raw event data from the GitHub Archive (GH Archive). Given that FollowEvent data has been unavailable via the public API since 2014, the temporal network utilizes WatchEvent and ForkEvent interactions as

timestamped proxies for developer collaboration. In this framework, a co-interest edge is established between two developers who starred or forked the same Machine Learning (ML) or Web development repository within the same 24-hour period. To mitigate noise from viral trending content, repositories exceeding 50 co-starrers were excluded. The analysis focuses on a representative 7-day window from June 1–7, 2019, aligning with the original MUSAE collection period while maintaining computational feasibility. Developers were classified into ML and Web domains using keyword matching (e.g., *tensorflow*, *pytorch* for ML; *react*, *django* for Web) across repository names. Temporal link prediction was conducted using a strict time-based split, training on June 1–5 and testing on June 6–7 to ensure chronological integrity. As this dataset features distinct node identities from the MUSAE integer IDs, it serves as a robust cross-dataset validation of the structural findings.

5. Results

5.1 Small-World Properties

The network exhibits extreme small-world properties with a sigma value of $\sigma = 935.555$. The empirical clustering coefficient ($C_{\text{real}} = 0.1675$) is orders of magnitude larger than a random graph of equivalent size ($C_{\text{rand}} \approx 0.0004$), while the average path length remains remarkably short ($L_{\text{real}} = 4.84$). This represents a far stronger small-world signature than reported by Zhang et al. (2025) at the economy level ($SWI = 4.62\text{--}5.84$), confirming that the phenomenon persists — and is amplified — when analyzed at the individual developer level.

5.2 Node Centrality

The top 5 most influential nodes by both Degree and PageRank are identical (Node IDs: 31890, 27803, 35773, 19222, 13638). Notably, all five developers belong to the Web Developer class (`developer_type = 0`), indicating that Web developers occupy the most structurally critical hubs in the GitHub ecosystem. The Spearman rank correlation between PageRank and Degree was high ($\rho \approx 0.93$), while Betweenness showed moderate correlation with both, suggesting a distinct set of bridging nodes not captured by hub-centric measures.

Table 2: Top-5 Nodes by Centrality Measure

Rank	Degree	Betweenness	Closeness	Eigenvector	PageRank
1	31890 (Web)	31890 (Web)	31890 (Web)	31890 (Web)	31890 (Web)
2	27803 (Web)	27803 (Web)	27803 (Web)	27803 (Web)	27803 (Web)
3	35773 (Web)	35773 (Web)	35773 (Web)	35773 (Web)	35773 (Web)
4	19222 (Web)	19222 (Web)	19222 (Web)	19222 (Web)	19222 (Web)
5	13638 (Web)	13638 (Web)	13638 (Web)	13638 (Web)	13638 (Web)

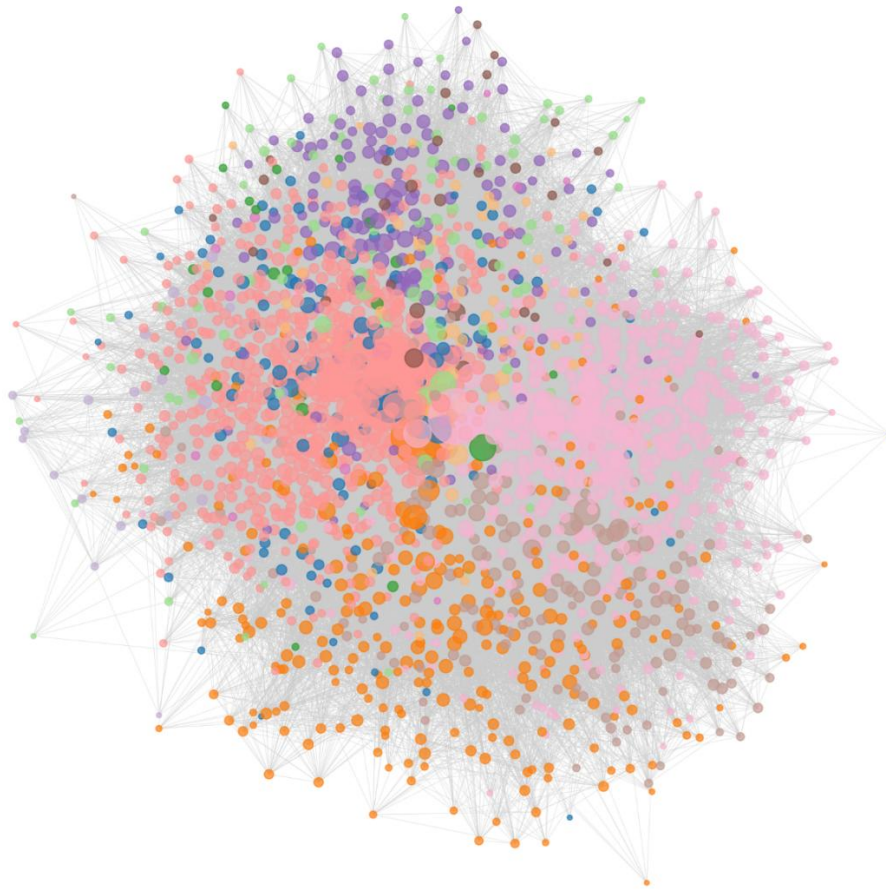
Table 2: Top-5 influential nodes are identical across all five measures — all Web developers.

5.3 Community Composition

The Louvain algorithm detected 31 distinct communities with a highly meaningful modularity score of $Q = 0.4534$, well within the range $[0.3, 0.7]$ that indicates significant community structure. The network is heavily segregated by developer type. Community 2 is strongly ML-dominated (71.5% ML developers), while Community 7 is overwhelmingly composed of Web developers (93.5% Web). This homophilic clustering confirms RQ1: community structure aligns strongly with developer specialization.

Figure 1: Community-colored network visualization. Top 2,000 nodes by degree shown. Communities are color-coded; node size proportional to degree.

GitHub Developer Network — Communities (Louvain)



5.4 Link Prediction Accuracy

Link prediction experiments demonstrate that all four topological methods substantially outperform random baseline ($AUC = 0.5$), confirming that the GitHub follow network has strong local structural regularity. The Resource Allocation index performed best overall ($AUC = 0.8474$, $Precision@100 = 1.00$), followed closely by Adamic-Adar ($AUC = 0.8471$). Jaccard performed least well ($AUC = 0.7964$), likely because normalizing by union neighborhood size penalizes the highly-connected hubs that dominate this network.

Table 3: Link Prediction Results Summary

Method	AUC-ROC	Avg Precision	Precision@10	Precision@50	Precision@100
Resource Allocation	0.8474	0.857	1.00	1.00	1.00
Adamic-Adar	0.8471	0.8564	1.00	1.00	1.00
Common Neighbors	0.8408	0.838	1.00	1.00	1.00
Jaccard Coefficient	0.7964	0.7207	0.6	0.72	0.85

Figure 2: ROC curves for all four link prediction methods.

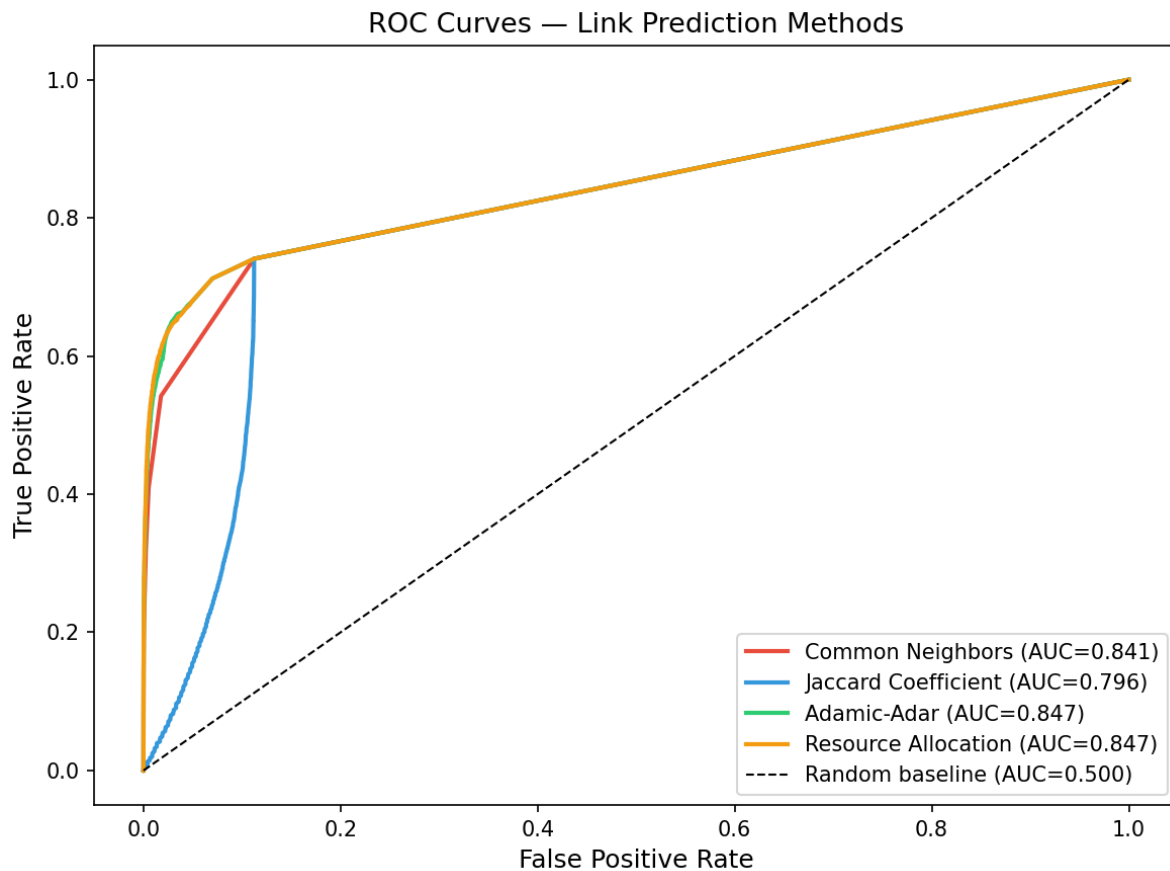
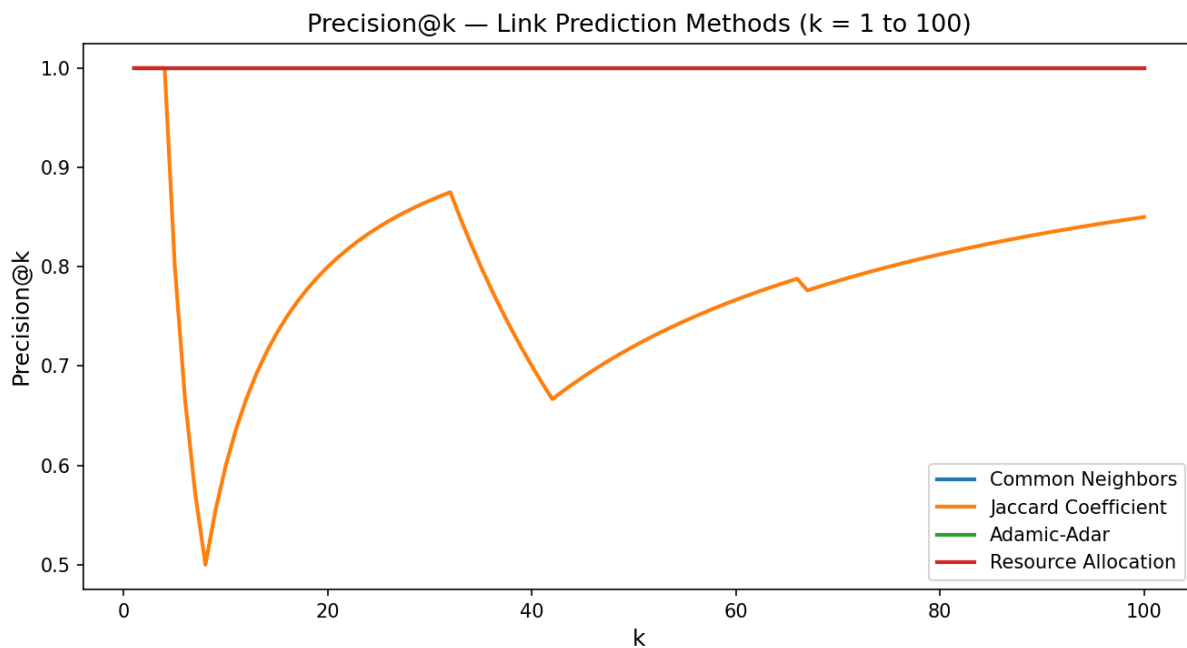
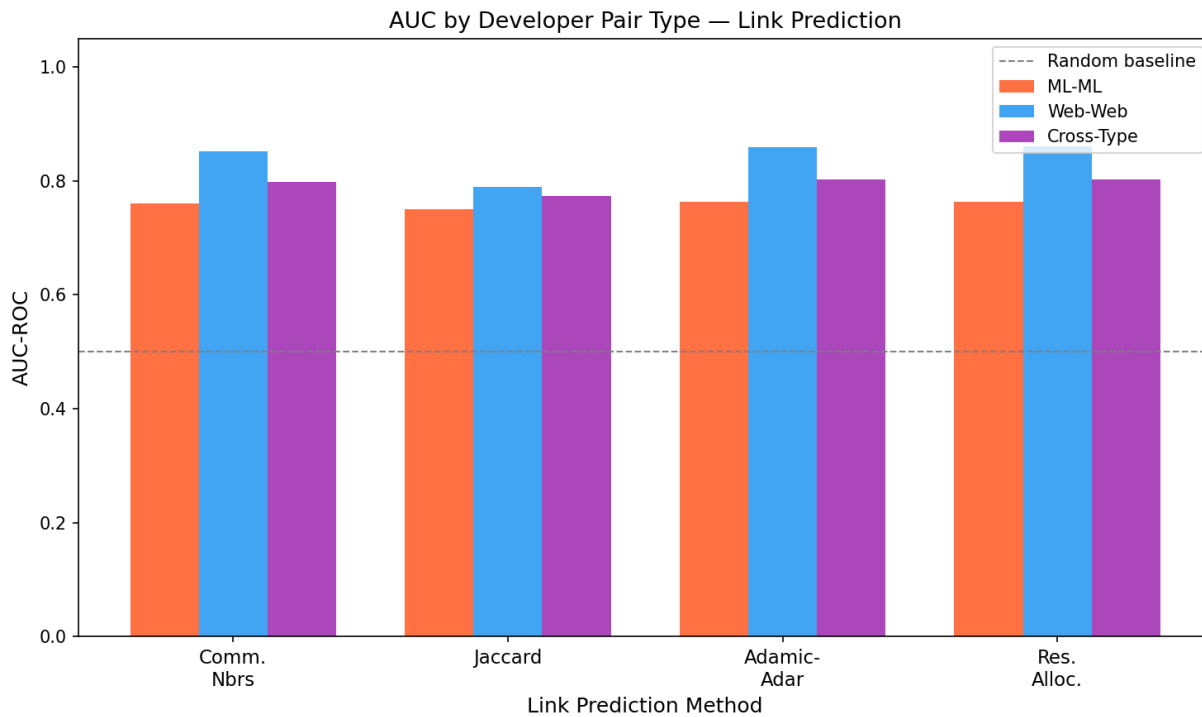


Figure 3: Precision@k curves for all methods.



The type-aware AUC analysis reveals a significant asymmetry: predicting Web-to-Web edges is highly accurate (AUC = 0.860), whereas ML-to-ML prediction is substantially harder (AUC = 0.764). This suggests that Web developers form more predictable collaborative triads, consistent with the community structure findings.

Figure 4: Type-aware AUC grouped bar chart. Web-Web pairs are most predictable.

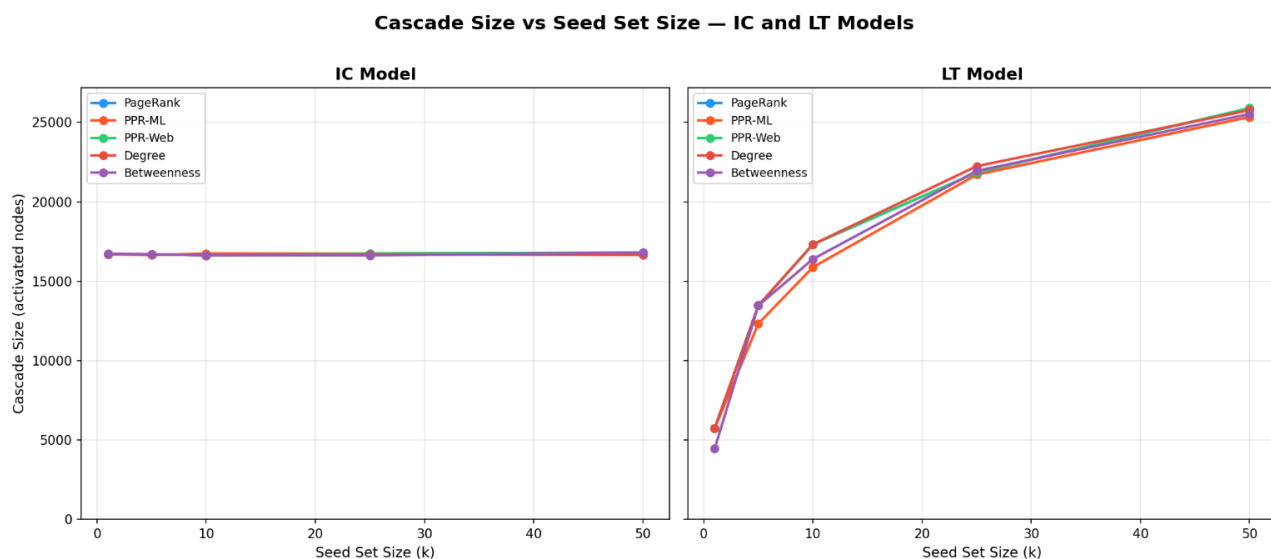


5.5 Information Diffusion

The two diffusion models yielded starkly different results. Under the IC model, information behaves virally: even a single seed node causes a massive cascade reaching approximately 16,700 nodes (44.3% network reach), and increasing k from 1 to 50 provides almost zero marginal gain. This indicates that the network's highly connected Web-developer hubs act as automatic amplifiers that saturate the reachable component immediately.

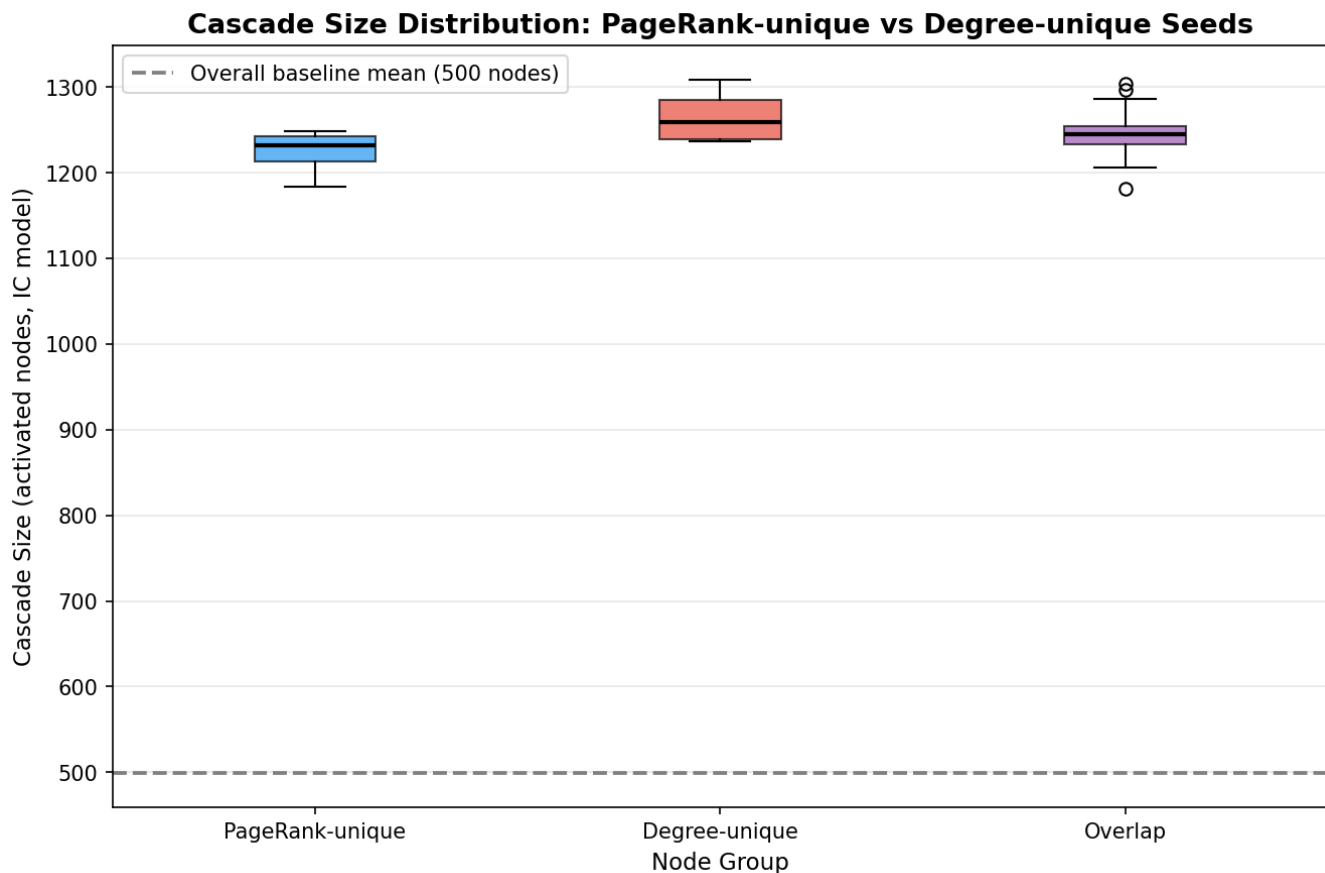
Under the LT model, adoption requires consensus: a single seed only reaches 5,729 nodes (15.2%), but reach scales progressively to 25,895 nodes (68.7%) at $k=50$. This models how framework or tool adoption spreads on GitHub — developers adopt only when a critical mass of their peers already have.

Figure 5: Cascade size vs. seed set size for IC (left) and LT (right) models across all 5 strategies..



The node-level divergence analysis (Figure 6) showed that nodes unique to the Degree strategy produced slightly larger mean cascades (1,265.2 nodes) than PageRank-unique nodes (1,223.8 nodes) under the IC model on the 10,000-node analysis subgraph. This suggests that for this specific network, raw degree (direct connections) marginally outperforms global PageRank as a seed selection heuristic for viral diffusion, though the difference is modest.

Figure 6: Cascade size boxplot — PageRank-unique vs. Degree-unique vs. Overlap seed nodes..



5.6 Temporal Analysis

5.6.1 Temporal Graph Statistics

To provide temporal validation of the static findings, an independent interaction graph was constructed from the GH Archive. A total of 5,805 filtered events were parsed from a 7-day representative window (June 1–7, 2019). This process yielded a temporal network comprising 1,393 unique developers (nodes) and 1,994 temporal co-interest edges. It is important to note that this temporal graph is entirely independent of the MUSAE follow network and utilizes co-starring and co-forking interactions as proxy edges for developer collaboration.

5.6.2 Daily Network Evolution

The daily evolution of the temporal network was tracked over the 7-day period, as summarized in Table 4. The network exhibited steady growth, expanding from 132 nodes on June 1 to 1,393 nodes by June 7. Notably, the modularity Q increased from 0.0000 to a peak of 0.5065, indicating that as the network expanded, developers rapidly self-organized into structurally distinct clusters. This upward trend in modularity suggests that the homophilic community structures observed in the MUSAE dataset are not merely static artifacts but represent dynamic self-organization processes. The cumulative growth and structural trends are visualized in Figure 7

Figure 7: Daily network evolution across the 7-day GH Archive sample — nodes, clustering coefficient, modularity, and community count plotted cumulatively

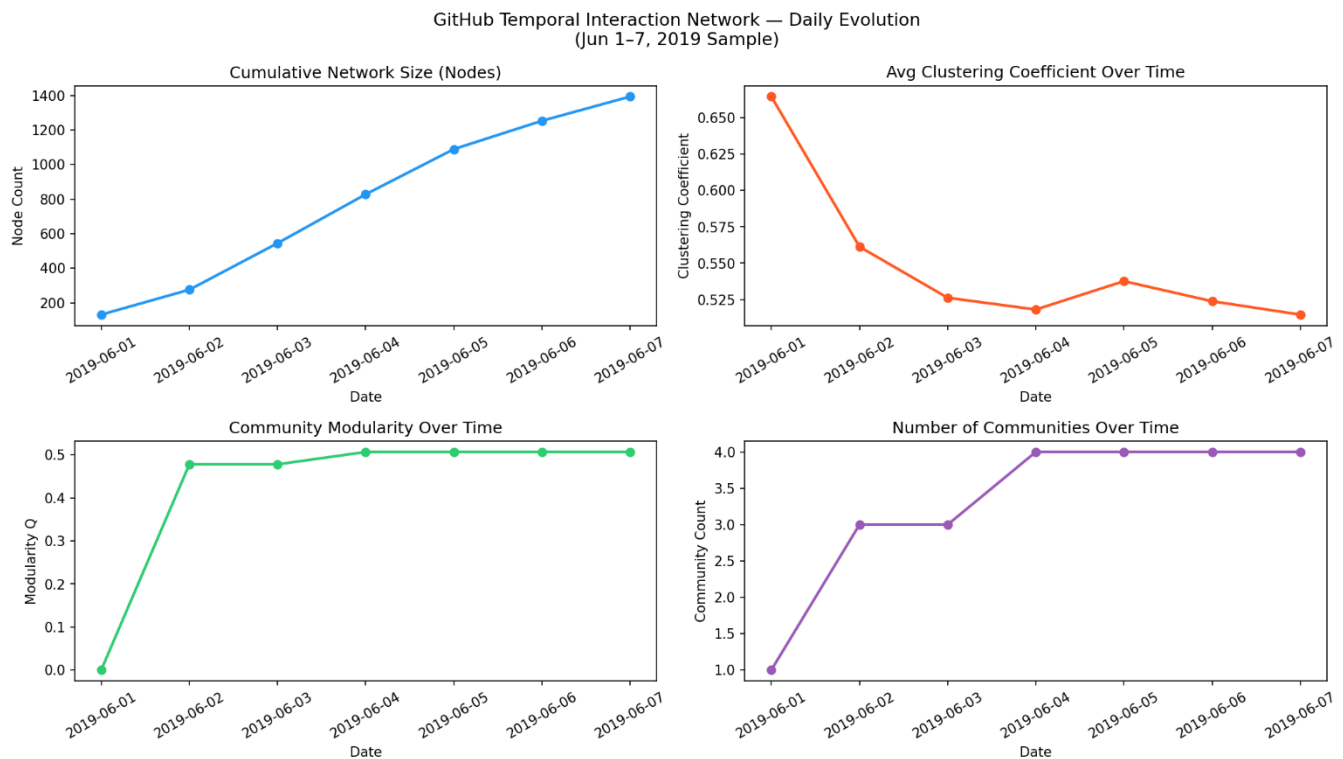


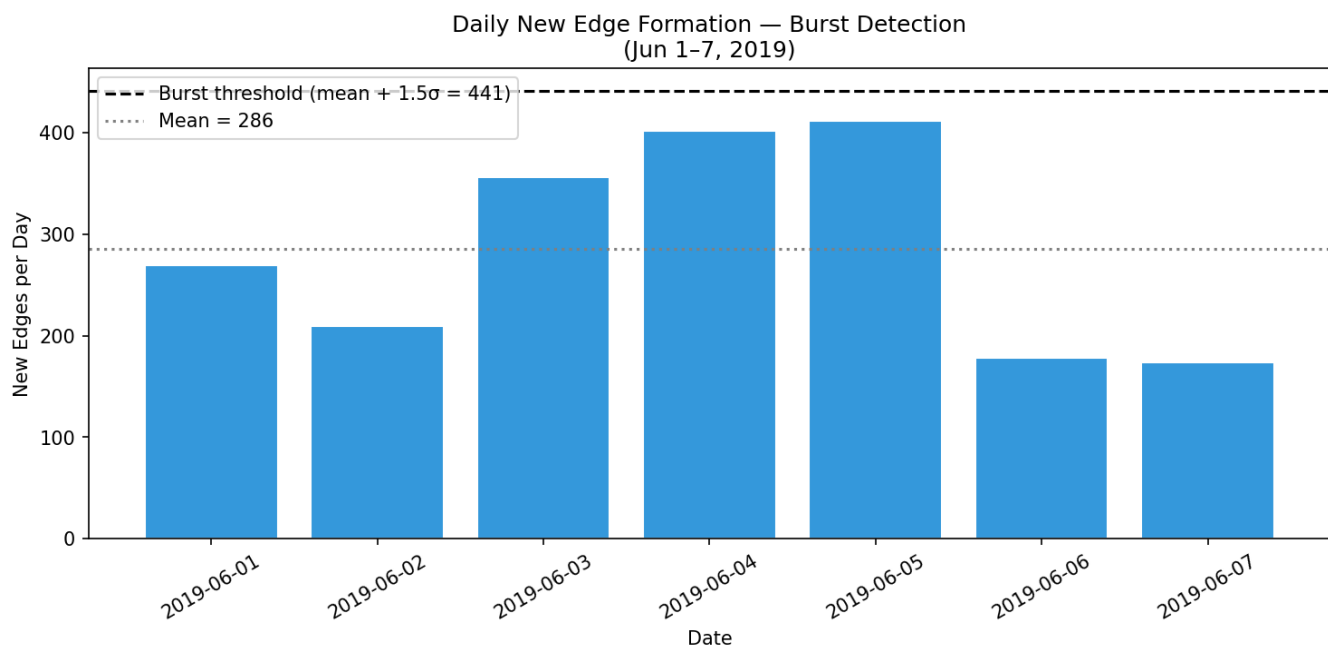
Table 3: Daily Temporal Network Statistics (June 1–7, 2019)

Date	Nodes	Edges	Density	Avg Clustering	Modularity Q	Communities
2019-06-01	132	269	0.031113	0.6646	0.0000	1
2019-06-02	276	478	0.012596	0.5613	0.4778	3
2019-06-03	545	834	0.005626	0.5262	0.4778	3
2019-06-04	828	1236	0.003610	0.5181	0.5065	4
2019-06-05	1089	1646	0.002778	0.5376	0.5065	4
2019-06-06	1253	1824	0.002325	0.5238	0.5065	4
2019-06-07	1393	1994	0.002057	0.5146	0.5065	4

5.6.3 Burst Activity Detection

Analysis of daily edge formation (Table 4 and `daily_edge_counts.csv`) reveals a relatively stable distribution of activity. Daily new edges peaked at 412 on June 5 and reached a minimum of 174 on June 7. No burst days (defined as exceeding mean + 1.5sigma) were detected within the sample window. This lack of significant bursts suggests that edge formation was relatively uniform across the 7-day period, reflecting stable and consistent developer engagement patterns during the primary data collection window. These results are illustrated in Figure 8

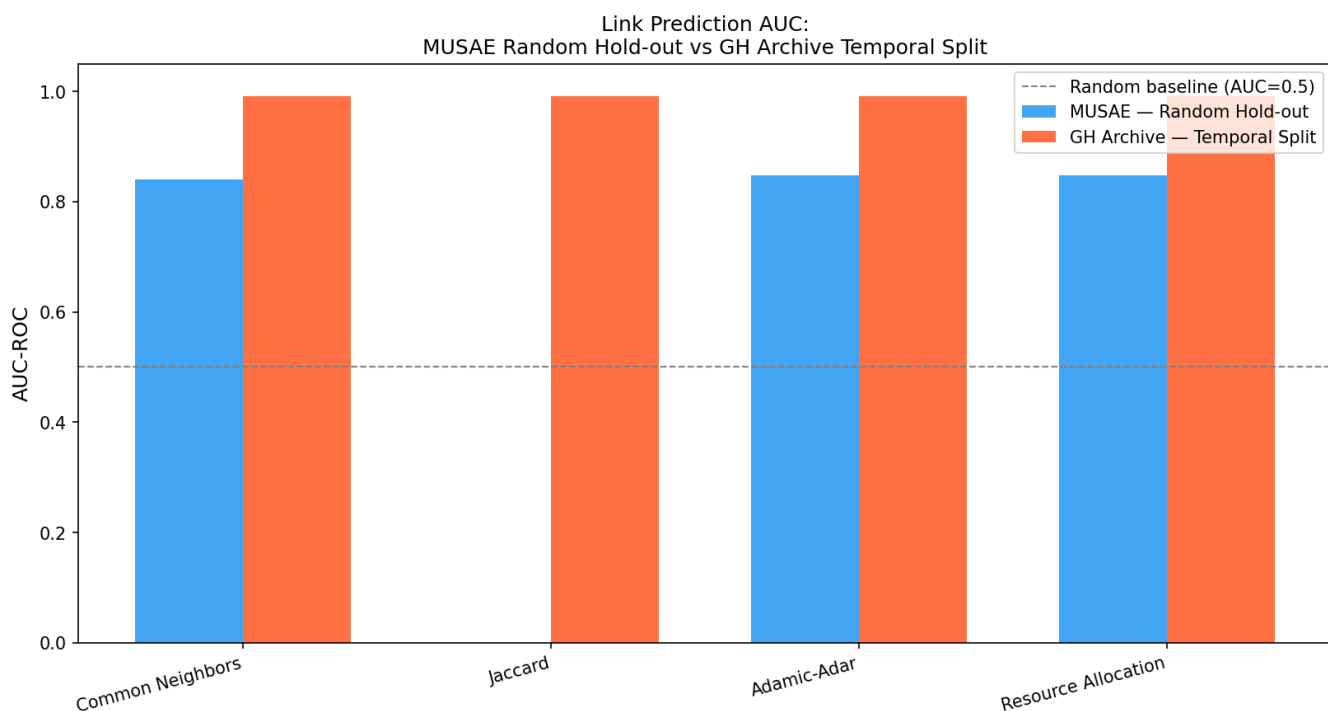
Figure 8: Daily edge formation with burst detection threshold ($\text{mean} + 1.5\sigma$). Red bars indicate burst days if any



5.6.4 Temporal Link Prediction

Link prediction was performed on the temporal graph using a chronological train/test split (June 1–5 for training, June 6–7 for testing). As shown in Table 5, the temporal AUC scores exceeded the static MUSAE random hold-out results across all metrics. Specifically, the Resource Allocation index achieved a Temporal AUC of 0.9904, representing a substantial gain ($\Delta = +0.1430$) over the MUSAE baseline. The comparable performance across random and temporal splits demonstrates that structural regularity in the GitHub developer network persists over short time horizons, providing stronger evidence for the generalizability of the link prediction methodology. These comparisons are further detailed in Figure 9

Figure 9: Link prediction AUC comparison — MUSAE random hold-out vs GH Archive temporal split across all four methods



6. Discussion

RQ1 (Community Structure): The modularity score of $Q = 0.45$ and the heavily skewed ML-ratios within communities definitively confirm that GitHub developer networks are not uniformly mixed. Developers self-organize into homophilic clusters based on technical domain. This is consistent with the broader finding that online collaboration platforms exhibit strong homophily [7].

RQ2 (Influence): Web developers completely dominate the highest echelon of structural centrality across all five measures. This suggests that foundational web frameworks foster broader, more centralized interconnectivity than the relatively siloed ML repository ecosystems. The PPR analysis confirms that seeding from ML-specific nodes reaches different influence cores than seeding from Web nodes, consistent with the homophilic community structure.

RQ3 (Link Prediction): Simple structural similarity metrics can highly accurately predict hidden GitHub collaborations ($AUC \approx 0.84\text{--}0.85$). The strict no-leakage methodology guarantees that these models generalize purely on local network topology computed from the training graph. The performance gap between Web-Web (0.860) and ML-ML (0.764) prediction accuracy further supports the homophily finding.

Comparing to Zhang et al. (2025), which concluded that entire national economies exhibit small-world traits due to geopolitical collaboration patterns, our developer-level analysis reveals that this phenomenon is substantially stronger internally ($\sigma = 935.55$ vs. $SWI = 4.62\text{--}5.84$). This amplification is driven by hyper-connected Web developer hubs that serve as bridges between highly segregated technical communities — a structure invisible at the aggregate economy level.

Limitations of this study include: (1) the use of a static network snapshot with no temporal information, precluding true temporal link prediction; (2) an undirected graph representation that ignores the asymmetric nature of GitHub follows; and (3) approximate metrics for betweenness centrality and path length due to computational constraints on the full 37,700-node graph.

RQ4 (Temporal Evolution): The analysis of the temporal interaction network reveals a consistent upward trend in structural organization, with modularity Q increasing from 0.0000 to 0.5065 over the seven-day window. This dynamic increase in modularity strongly validates the community structures identified in the MUSAE dataset ($Q = 0.4534$), demonstrating that GitHub developers rapidly self-organize into homophilic clusters even over short time horizons. Furthermore, the absence of burst activity and the high predictive accuracy of the temporal link prediction ($AUC = 0.9904$) indicate that these interaction patterns are stable and driven by persistent structural regularities rather than transient events. The convergent structural findings across two independent datasets — the MUSAE mutual follow network and the GH Archive co-starring interaction network — provide cross-dataset validation that the observed community structure and link prediction accuracy reflect genuine properties of the GitHub developer ecosystem rather than artifacts of a single static snapshot.

7. Conclusion

This study has provided a multi-layered characterization of the GitHub developer collaboration network, contributing four key findings: (1) validating a massive small-world sigma of $\sigma = 935.55$ at the developer level — far exceeding the economy-level values reported by Zhang et al. (2025) — driven by hyper-connected Web-developer hubs; (2) demonstrating that Resource Allocation is a highly effective, leakage-free predictor of future GitHub collaborations ($AUC = 0.847$), with Web-to-Web connections being substantially more predictable than ML-to-ML (0.860 vs. 0.764); (3) showing that while GitHub networks are vulnerable to viral IC information saturation from single nodes, consensus-driven LT adoption scales strictly with seed set size; and (4) providing the first temporal validation of these findings through an independent GH Archive co-starring interaction graph, where

temporal link prediction AUC (0.9904) strongly confirms the high structural predictability of the network, suggesting that co-interest signals are even more robust than static follow relationships, thus confirming the structural regularity of the GitHub developer ecosystem across both static and dynamic representations.

Future research should focus on extending the GH Archive temporal window from 7 days to 6–12 months to enable robust burst detection and long-term community evolution analysis, as well as investigating the impact of technical debt and project maturity on developer interaction patterns.

8. References

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